Project: Worked examples
Title: Thick-walled cylinder stresses (Lamé)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Radial, hoop and axial stresses at the bore and the outer surface of a thick-walled cylinder under internal and external pressure, the von Mises and Tresca equivalents, utilisation against an allowable, and the thin-wall estimate for comparison.
$A = (a^2 p_i - b^2 p_o)/(b^2 - a^2)$, $B = a^2 b^2 (p_i - p_o)/(b^2 - a^2)$; $\sigma_r = A - B/r^2$, $\sigma_\theta = A + B/r^2$; $\sigma_z = A$ for closed ends, 0 for open. von Mises from the three principal stresses; Tresca = max principal difference.
Ref: Roark's Formulas for Stress and Strain Ch. 13, Table 13.5 — Thick-walled cylinders
Ref: Shigley's Mechanical Engineering Design §3-14 — Stresses in pressurised cylinders

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
a	Inner radius	50	mm	drawing: DRG-CYL-01 rev A	0.1 … 5000 mm
b	Outer radius	100	mm	drawing: DRG-CYL-01 rev A	0.1 … 5000 mm
p_i	Internal pressure	50	MPa	spec: SPEC-PV-2 §3 design pressure	0 … 1000 MPa
p_o	External pressure	0	MPa	assumption: Atmospheric outside.	0 … 1000 MPa
closed	Closed ends (axial stress carried by the wall)	yes		drawing: DRG-CYL-01 end caps rev A	any
sigma_allow	Allowable stress	200	MPa	spec: SPEC-PV-2 §5	1 … 3000 MPa

3. Assumptions

A1. Long cylinder, elastic, isotropic; stresses vary with radius only. (Lamé solution; end effects neglected.)
A2. Closed ends carry the pressure end load as a uniform axial stress. (σ_z = A.)
A3. p_o (External pressure) is taken as 0 MPa. (Atmospheric outside.)

4. Working

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[7] A = (a^2 * p_i - b^2 * p_o) / (b^2 - a^2)                    Roark Table 13.5
= ((50 mm)^2 × 50 MPa - (100 mm)^2 × 0 MPa) / ((100 mm)^2 - (50 mm)^2)
= 16.667 MPa
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Lamé constant A

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[8] B = a^2 * b^2 * (p_i - p_o) / (b^2 - a^2)                    Roark Table 13.5
= (50 mm)^2 × (100 mm)^2 × (50 MPa - 0 MPa) / ((100 mm)^2 - (50 mm)^2)
= 166670 MPa·mm²
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Lamé constant B

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[9] sigma_r_a = A - B / a^2
= 16.667 MPa - 166670 MPa·mm² / (50 mm)^2
= -50 MPa
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Radial stress at the bore

[10] sigma_t_a = A + B / a^2
= 16.667 MPa + 166670 MPa·mm² / (50 mm)^2
= 83.333 MPa
Hoop stress at the bore
[11] sigma_r_b = A - B / b^2
= 16.667 MPa - 166670 MPa·mm² / (100 mm)^2
= 1.8626e-15 MPa
Radial stress at the outer surface
[12] sigma_t_b = A + B / b^2
= 16.667 MPa + 166670 MPa·mm² / (100 mm)^2
= 33.333 MPa
Hoop stress at the outer surface
[13] sigma_z = A
= 16.667 MPa
= 16.667 MPa
Axial stress (closed ends)
[14] sigma_vm_a = sqrt(((sigma_t_a - sigma_r_a)^2 + (sigma_r_a - sigma_z)^2 + (sigma_z - sigm
2)
= sqrt(((83.333 MPa - (-50 MPa))^2 + ((-50 MPa) - 16.667 MPa)^2 + (16.667 MPa - 83.333 MPa)^2
= 115.47 MPa
von Mises stress at the bore
[15] sigma_vm_b = sqrt(((sigma_t_b - sigma_r_b)^2 + (sigma_r_b - sigma_z)^2 + (sigma_z - sigm
2)
= sqrt(((33.333 MPa - 1.8626e-15 MPa)^2 + (1.8626e-15 MPa - 16.667 MPa)^2 + (16.667 MPa - 33.
MPa)^2) / 2)
= 28.868 MPa
von Mises stress at the outer surface
[16] sigma_tr_a = max(sigma_t_a, sigma_r_a, sigma_z) - min(sigma_t_a, sigma_r_a, sigma_z)
= max(83.333 MPa, -50 MPa, 16.667 MPa) - min(83.333 MPa, -50 MPa, 16.667 MPa)
= 133.33 MPa
Tresca stress at the bore
[17] sigma_tr_b = max(sigma_t_b, sigma_r_b, sigma_z) - min(sigma_t_b, sigma_r_b, sigma_z)
= max(33.333 MPa, 1.8626e-15 MPa, 16.667 MPa) - min(33.333 MPa, 1.8626e-15 MPa, 16.667 MPa)
= 33.333 MPa
Tresca stress at the outer surface
[18] sigma_vm_max = max(sigma_vm_a, sigma_vm_b)
= max(115.47 MPa, 28.868 MPa)
= 115.47 MPa
Maximum von Mises stress
[19] u = sigma_vm_max / sigma_allow
= 115.47 MPa / 200 MPa
= 0.57735
Utilisation
[20] sigma_thin = (p_i - p_o) * (a + b) / 2 / (b - a)
= (50 MPa - 0 MPa) × (50 mm + 100 mm) / 2 / (100 mm - 50 mm)
= 75 MPa
Thin-wall hoop estimate at the mean radius
[21] k_wall = b / a
= 100 mm / 50 mm
= 2
Wall ratio b/a

5. Results

SYMBOL	VALUE	DESCRIPTION

SYMBOL	VALUE	DESCRIPTION
sigma_r_a	-50 MPa	Radial stress, bore
sigma_t_a	83.333 MPa	Hoop stress, bore
sigma_r_b	1.8626e-15 MPa	Radial stress, outer
sigma_t_b	33.333 MPa	Hoop stress, outer
sigma_z	16.667 MPa	Axial stress
sigma_vm_a	115.47 MPa	von Mises, bore
sigma_vm_b	28.868 MPa	von Mises, outer
sigma_tr_a	133.33 MPa	Tresca, bore
sigma_tr_b	33.333 MPa	Tresca, outer
sigma_vm_max	115.47 MPa	Maximum von Mises
u	0.57735	Utilisation
sigma_thin	75 MPa	Thin-wall hoop estimate
k_wall	2	Wall ratio

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	von Mises against allowable	u = 0.57735	u ≤ 1	0.577	

7. Applicability

No warnings.

8. Verification

Template mech.thick-cylinder@2.0.0 Content hash d23e76ef30d008ef Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0